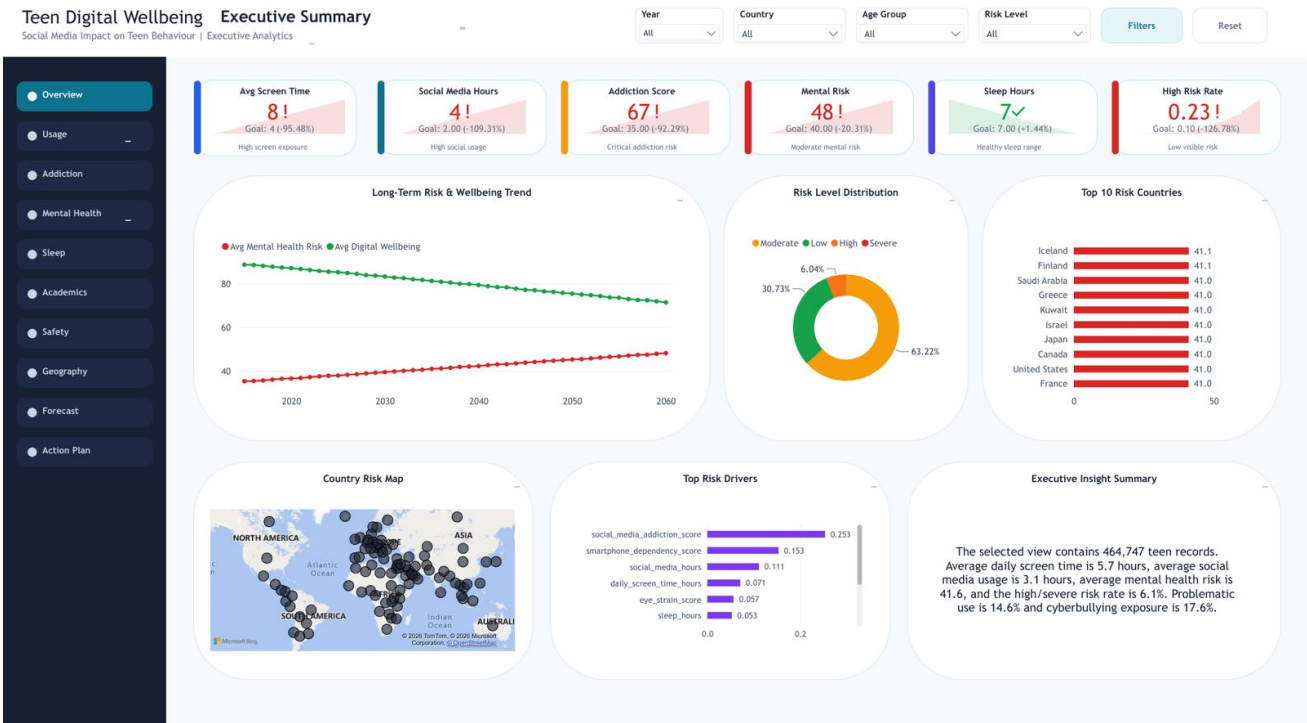


Teen Digital Wellbeing Analytics

Social Media Impact on Teen Behaviour (2015-2060)

Power BI Service Deployment, Semantic Model, Report Pages, App Distribution, and Team Handover Documentation



Cover preview: Executive Summary report page from the uploaded Power BI PDF export

Document version: 1.1 | Prepared by: Prakhar Tated | Report state: Pages 1-10 | Updated: Expanded DAX measure catalog added

Document Control

Field	Details
Project	Teen Digital Wellbeing Analytics
Report	Social Media Impact on Teen Behaviour (2015-2060)
Platform	Power BI Desktop, Power BI Service, Deployment Pipelines, Power BI App
Audience	BI team, QA team, project stakeholders, deployment owners
Included pages	01 Executive Summary through 09 Forecast Analysis, plus 12 Action Plan
Source screenshots	Uploaded Power BI PDF export containing 10 report pages

Table of Contents

- 1. Executive Overview
- 2. Solution Scope and Page Inventory
- 3. Data Sources and Semantic Model Design
- 4. Power Query and Data Preparation
- 5. Core DAX Measures and Calculated Columns
- 6. Report Page Documentation
- 7. Power BI Service Deployment Runbook
- 8. Deployment Pipeline: DEV to STAGE to PROD
- 9. Power BI App Distribution
- 10. QA, Validation, and Troubleshooting
- 11. Maintenance and Ownership
- 12. Appendix: Key Measures and Operational Checklists
- 13. Appendix: Expanded DAX Measure Catalog

1. Executive Overview

This document provides a complete handover reference for the Teen Digital Wellbeing Analytics Power BI solution. It explains the business purpose, report pages, semantic model, data preparation approach, DAX logic, deployment pipeline, app distribution process, QA checklist, and troubleshooting steps.

The report analyzes social media impact on teen behaviour across usage patterns, addiction dependency, mental health, sleep and physical wellbeing, academic impact, online safety, geography risk, forecast analysis, and action planning.

Business Area	What the dashboard answers
Executive analytics	What is the overall teen digital wellbeing risk and where should leadership focus first?
Usage behaviour	How much time are teens spending on screens, social media, and platforms?
Addiction and dependency	Which groups show higher addiction, smartphone dependency, FOMO, and problematic use?
Mental health	How are anxiety, depression, stress, loneliness, and mental risk changing?
Sleep and wellbeing	How do screen habits relate to sleep, activity, eye strain, headaches, and physical risk?
Academics	How are screen time and social media linked to concentration, GPA, attendance, and homework?
Online safety	What is the exposure to cyberbullying, harmful content, misinformation, and privacy risks?
Geography	Which countries, regions, continents, and income groups have the highest risk?
Forecast	How are addiction and mental health risks expected to change in 2030-2060?
Action plan	Which groups require immediate intervention, monitoring, or preventive support?

2. Solution Scope and Page Inventory

The production dashboard should include only the following report pages. Pages 10 and 11 from earlier build iterations are intentionally excluded from the final deployment scope.

Page	Page Name	Primary KPI Row	Key Visuals
01	Executive Summary	Avg Screen Time, Social Media Hours, Addiction Score, Mental Risk, Sleep Hours, High Risk Rate	Long-Term Risk & Wellbeing Trend, Risk Level Distribution, Top 10 Risk Countries...
02	Usage Analysis	Avg Screen Time, Social Media Hours, Night Screen Time, Daily Notifications	Screen Time Trend, Usage by Age Group, Platform Usage Ranking...
03	Addiction Dependency	Addiction Score, Smartphone Dependency, FOMO Score, Problematic Use Rate	Addiction Trend, Addiction by Age Group, Dependency by Country...
04	Mental Health	Mental Health Risk, Anxiety Score, Depression Score, Stress Score	Mental Health Trend, Risk by Age Group, Top Countries by Mental Risk...
05	Sleep & Physical Wellbeing	Sleep Hours, Physical Activity, Eye Strain, Physical Health Risk	Sleep & Physical Trend, Sleep by Age Group, Top Countries by Physical Risk...
06	Academic Impact	Academic Risk, GPA Score, Concentration, Homework Completion	Academic Trend, Risk by Age Group, Top Countries by Academic Risk...
07	Online Safety	Cyberbullying Exposure, Online Harassment, Harmful Content, Privacy Risk	Online Safety Trend, Safety by Age Group, Top Countries by Cyberbullying...
08	Geography Risk	Overall Risk, High Risk Rate, Digital Wellbeing, Cyberbullying Rate	Country Risk Map, Top Risk Countries, Risk by Region...
09	Forecast Analysis	Addiction Risk 2060, Mental Risk 2060, Addiction Change, Mental Change	Forecast Risk Trend, Top Countries by 2060 Risk, Forecast by Region...
10	Action Plan	Immediate Intervention, High Priority Monitoring, Actionable Group, Avg Action Risk	Action Priority Distribution, Top Intervention Countries, Intervention Area Breakdown...

3. Data Sources and Semantic Model Design

The solution uses a star-schema-oriented semantic model. The main behavioral records table is treated as the primary fact table. Supporting fact tables provide platform usage, forecast values, feature importance, and summary/risk profiles.

Table	Type	Purpose
FactTeenBehaviour	Fact	Main teen-level behavior and risk table. Used by most report pages.
FactPlatformUsage	Fact	Unpivoted platform usage hours by platform and

		demographic/geography fields.
FactForecastLong	Fact	Long-format forecast table for projected addiction and mental health risk.
FactFeatureImportance	Fact / static driver table	Model-level feature importance used for top risk driver visuals.
FactCountryYearSummary	Fact summary	Country-year aggregated risk and wellbeing values.
FactCountryRisk2026	Fact summary	Country risk profile snapshot.
FactRegionRisk2026	Fact summary	Regional risk profile snapshot.
DimDate	Dimension	Date/year dimension for historic trend pages.
DimForecastYear	Dimension	Forecast year dimension for page 09.
DimCountry	Dimension	Country, continent, region, and income group hub.
DimDemographic	Dimension	Age group and gender grouping.
DimIncomeGroup	Dimension	Income group and sort order.
DimRegion	Dimension	Region grouping using region key to avoid duplicates.
DimRisk	Dimension	Risk level ordering and slicer support.
DimScreenCategory / DimSleepCategory	Dimension	Screen and sleep categories for analysis and sorting.
DimUrbanRural	Dimension	Urban/rural classification.
Relationship principle		Implementation note
Star schema		Use dimensions to filter facts; avoid direct fact-to-fact relationships.
Cross-filter direction		Use single direction wherever possible to avoid ambiguity and performance issues.
Forecast model		Use DimForecastYear for FactForecastLong. Do not connect DimDate directly to FactForecastLong.
Region key		Use a composite region key such as continent region when DimRegion region names are not unique.
Primary platform analysis		Use FactTeenBehaviour[primary_platform] or DimPrimaryPlatform for addiction-by-platform visuals. Do not use FactPlatformUsage platform to filter addiction measures.

4. Power Query and Data Preparation

Preparation Area	Recommended action
Main fact load	Load the main synthetic teen behavior CSV as FactTeenBehaviour. Verify data types for year, numeric scores, flags, and category columns.
Platform unpivot	Create FactPlatformUsage by referencing FactTeenBehaviour, keeping demographic/geography columns and unpivoting platform-hour columns into platform and platform_hours.
Forecast long table	Load or transform forecast data into FactForecastLong with country, region, income_group, forecast_year, risk_type, projected_risk.
Category labels	Create row-level labels for problematic use, high mental risk, academic risk, cyberbullying, harmful content, and action priorities.
Column audit	Remove or hide columns that are not used in reporting, especially constant technical columns. Keep calculated labels and sort columns where needed.
Data types	Ensure forecast_year and year are whole numbers; projected_risk and score fields are decimal numbers; risk_type and category fields are text.

5. Core DAX Measures and Calculated Columns

Measures should be stored in a dedicated _Measures table to keep the field list clean and support team maintenance.

```

_Measures =
DATATABLE (
    "Measure Group", STRING,
    { { "Executive Summary Measures" } }
)

```

Measure Group	Representative Measures
Core record and score measures	Teen Records, Avg Screen Time Hours, Avg Social Media Hours, Avg Addiction Score, Avg Mental Health Risk, Avg Sleep Hours, Avg Overall Risk
Risk-rate measures	High Severe Risk Rate, Problematic Use Rate, Cyberbullying Exposure Rate, High Academic Risk Rate
Page 09 forecast measures	Projected Risk, Projected Addiction Risk, Projected Mental Health Risk, Projected Addiction Risk 2060, Forecast Change measures
Page 10 action measures	Immediate Intervention Rate, High Priority Monitoring Rate, Actionable Rate, Avg Action Risk
Status measures	KPI status text measures such as Within target, Above target, Critical risk, Manageable intervention load

Teen Digital Wellbeing Analytics - Team Documentation

```
Teen Records = COUNTROWS ( FactTeenBehaviour )

Avg Screen Time Hours = AVERAGE ( FactTeenBehaviour[daily_screen_time_hours] )
Avg Social Media Hours = AVERAGE ( FactTeenBehaviour[social_media_hours] )
Avg Addiction Score = AVERAGE ( FactTeenBehaviour[social_media_addiction_score] )
Avg Mental Health Risk = AVERAGE ( FactTeenBehaviour[mental_health_risk_score] )
Avg Overall Risk = AVERAGE ( FactTeenBehaviour[overall_risk_score] )

High Severe Risk Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[risk_level] IN { "High", "Severe" }
)

High Severe Risk Rate = DIVIDE ( [High Severe Risk Records], [Teen Records] )

Projected Risk = AVERAGE ( FactForecastLong[projected_risk] )

Projected Addiction Risk =
CALCULATE (
    [Projected Risk],
    FactForecastLong[risk_type] = "Addiction Risk"
)

Projected Addiction Risk 2060 =
CALCULATE (
    [Projected Addiction Risk],
    FactForecastLong[forecast_year] = 2060
)

Action Priority =
SWITCH (
    TRUE (),
    FactTeenBehaviour[risk_level] = "Severe"
        || FactTeenBehaviour[overall_risk_score] >= 70, "Immediate Intervention",
    FactTeenBehaviour[risk_level] = "High"
        || FactTeenBehaviour[overall_risk_score] >= 55, "High Priority Monitoring",
    FactTeenBehaviour[risk_level] = "Medium"
        || FactTeenBehaviour[overall_risk_score] >= 40, "Preventive Support",
    "General Awareness"
)
```

6. Report Page Documentation

Each section below includes the page objective, KPI row, key visuals, business use, and screenshot from the uploaded PDF export.

01 Executive Summary

Provides leadership-level overview of digital wellbeing performance, high-risk groups, risk distribution, country map, and top modeled risk drivers.

Item	Details
KPI row	Avg Screen Time, Social Media Hours, Addiction Score, Mental Risk, Sleep Hours, High Risk Rate
Key visuals	Long-Term Risk & Wellbeing Trend, Risk Level Distribution, Top 10 Risk Countries, Country Risk Map, Top Risk Drivers, Executive Insight Summary
Business use	Use this as the first review page for stakeholder walkthroughs and monthly executive reviews.

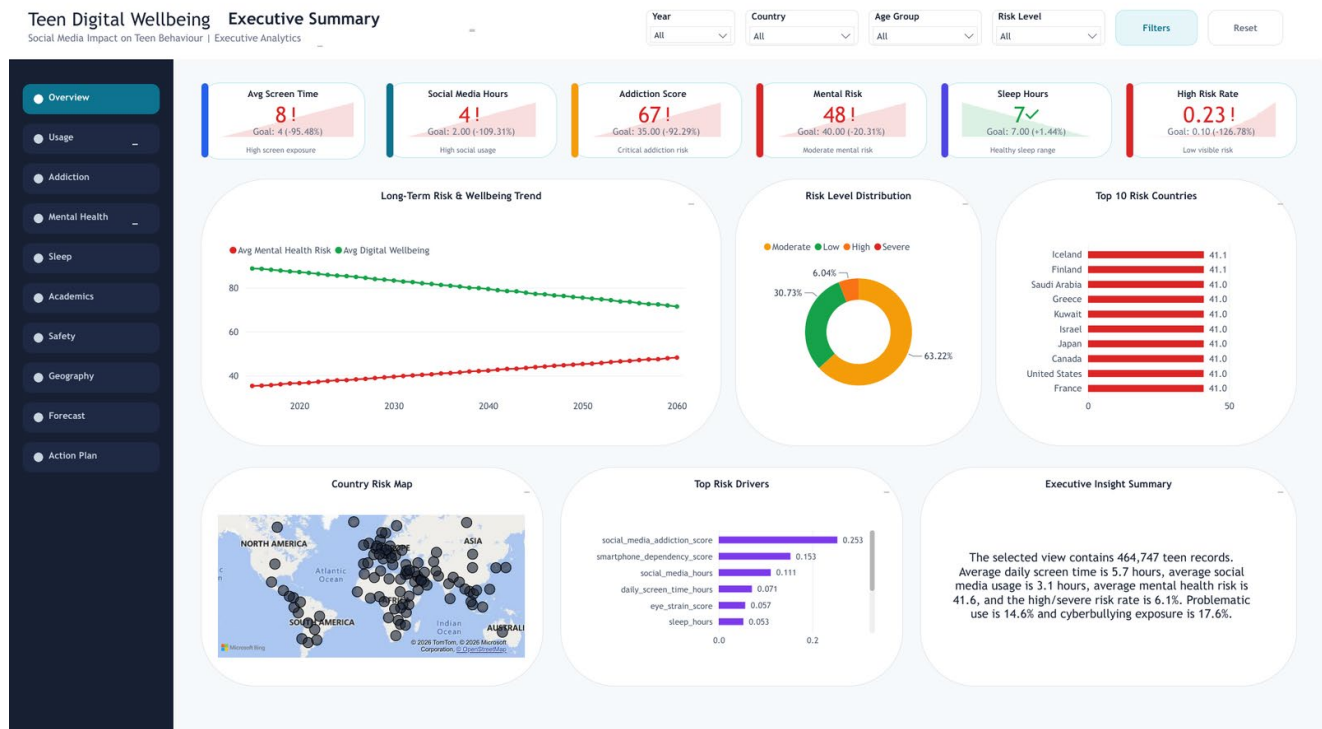


Figure 1: 01 Executive Summary page screenshot

02 Usage Analysis

Explains screen-time behavior across years, age groups, platforms, screen-time categories, and gender.

Item	Details
KPI row	Avg Screen Time, Social Media Hours, Night Screen Time, Daily Notifications
Key visuals	Screen Time Trend, Usage by Age Group, Platform Usage Ranking, Screen Time Category Distribution, Usage by Gender, Usage Insight Summary
Business use	Use this page to identify usage intensity and platform engagement patterns that may require education or coaching.

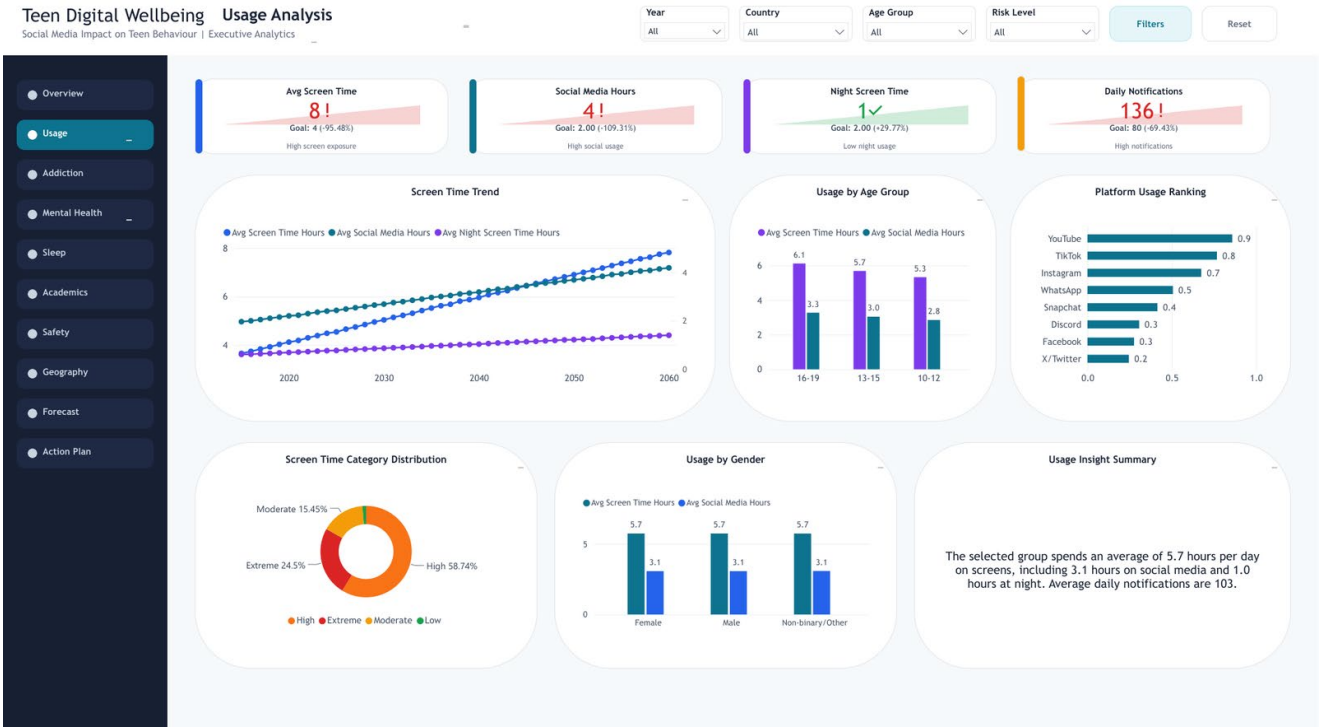
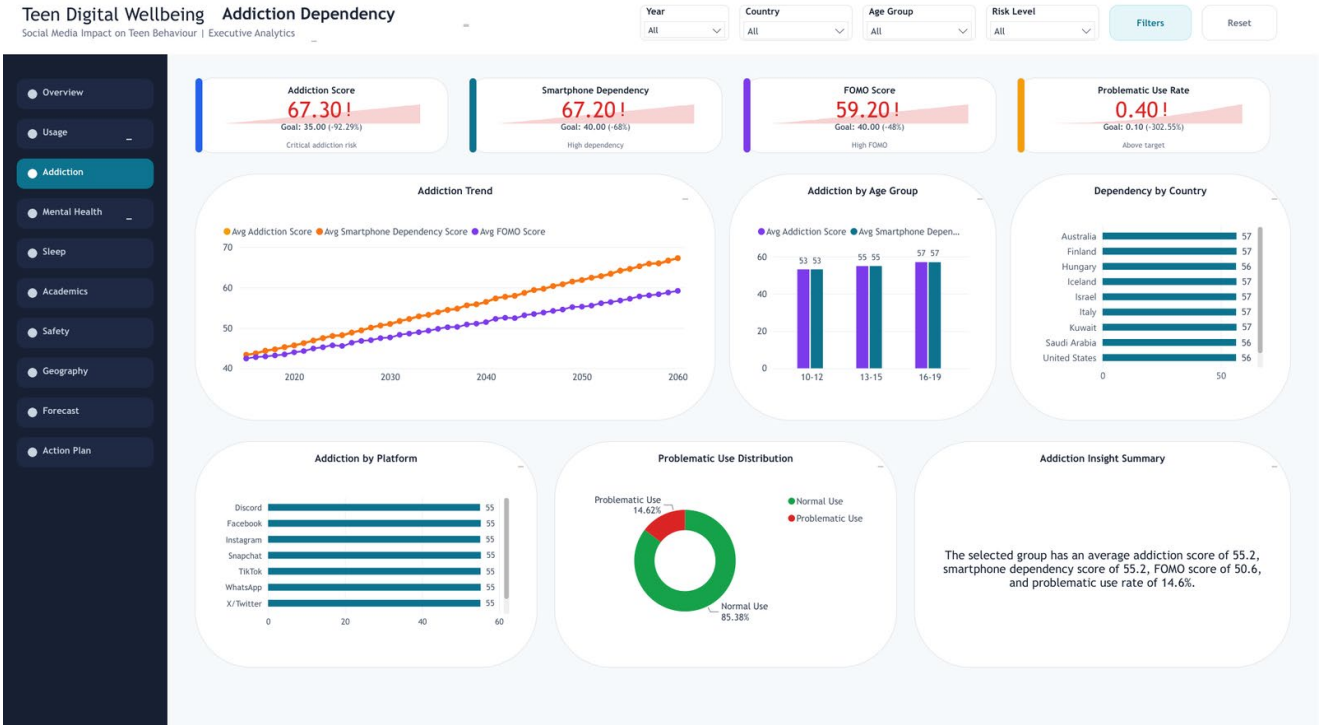


Figure 2: 02 Usage Analysis page screenshot

03 Addiction Dependency

Tracks addiction risk, smartphone dependency, FOMO, problematic use, and platform-level addiction indicators.

Item	Details
KPI row	Addiction Score, Smartphone Dependency, FOMO Score, Problematic Use Rate
Key visuals	Addiction Trend, Addiction by Age Group, Dependency by Country, Addiction by Platform, Problematic Use Distribution, Addiction Insight Summary
Business use	Use this page for intervention planning around compulsive usage and dependency patterns.



04 Mental Health

Analyzes mental health risk, anxiety, depression, stress, loneliness, and high mental-risk segments.

Item	Details
KPI row	Mental Health Risk, Anxiety Score, Depression Score, Stress Score
Key visuals	Mental Health Trend, Risk by Age Group, Top Countries by Mental Risk, Anxiety vs Depression, High Mental Risk Distribution, Mental Health Summary
Business use	Use this page to support mental health program planning and monitor changing risk scores over time.

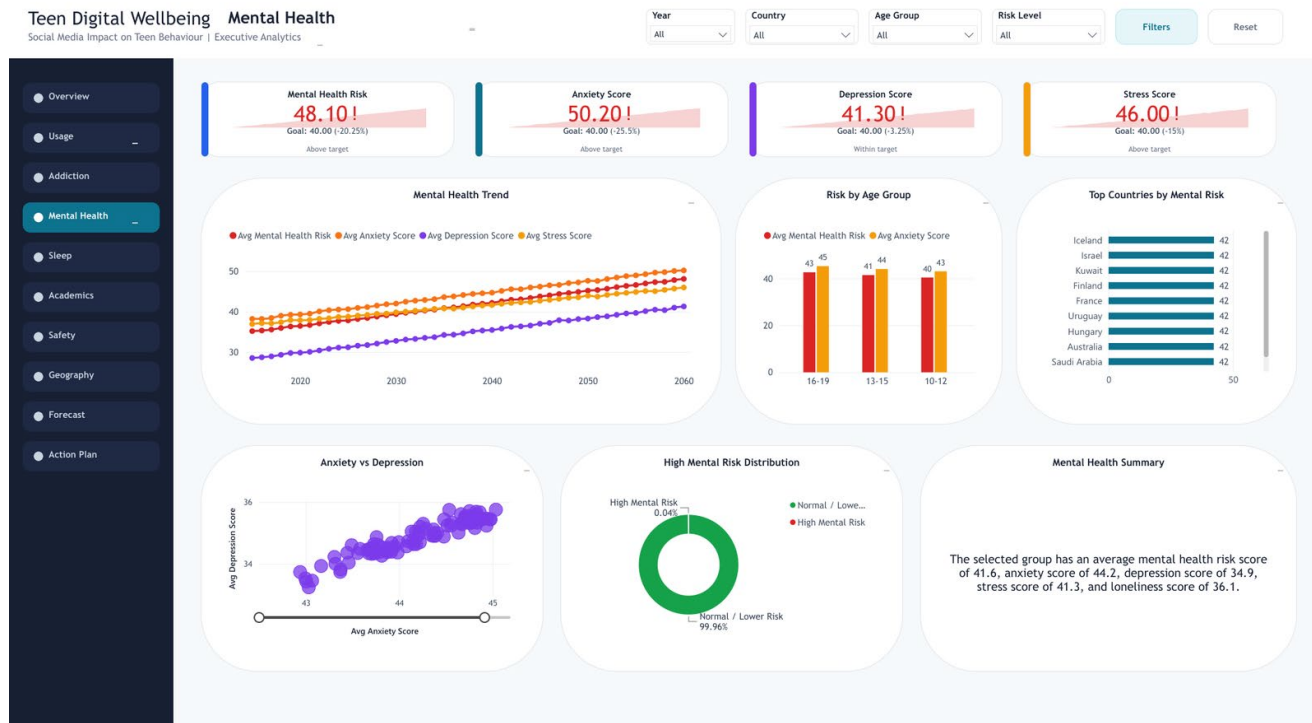


Figure 4: 04 Mental Health page screenshot

05 Sleep & Physical Wellbeing

Links screen behavior to sleep, physical activity, eye strain, headaches, and physical health risk.

Item	Details
KPI row	Sleep Hours, Physical Activity, Eye Strain, Physical Health Risk
Key visuals	Sleep & Physical Trend, Sleep by Age Group, Top Countries by Physical Risk, Sleep Category Distribution, Activity vs Screen Time, Physical Wellbeing Summary
Business use	Use this page for wellness and screen-time reduction initiatives.

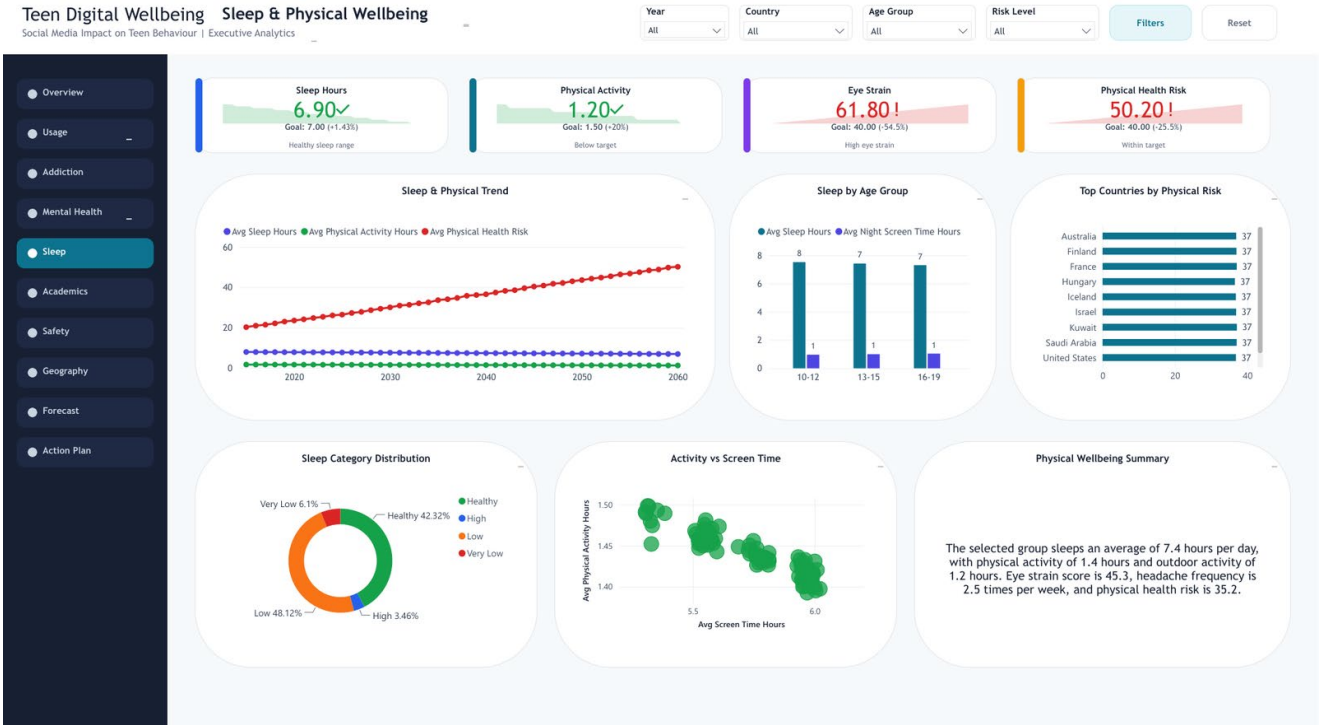


Figure 5: 05 Sleep & Physical Wellbeing page screenshot

06 Academic Impact

Shows academic risk, GPA, concentration, homework completion, attendance and screen-time academic impact.

Item	Details
KPI row	Academic Risk, GPA Score, Concentration, Homework Completion
Key visuals	Academic Trend, Risk by Age Group, Top Countries by Academic Risk, Screen Time vs GPA, Academic Risk Distribution, Academic Impact Summary
Business use	Use this page to identify the relationship between digital usage and academic outcomes.

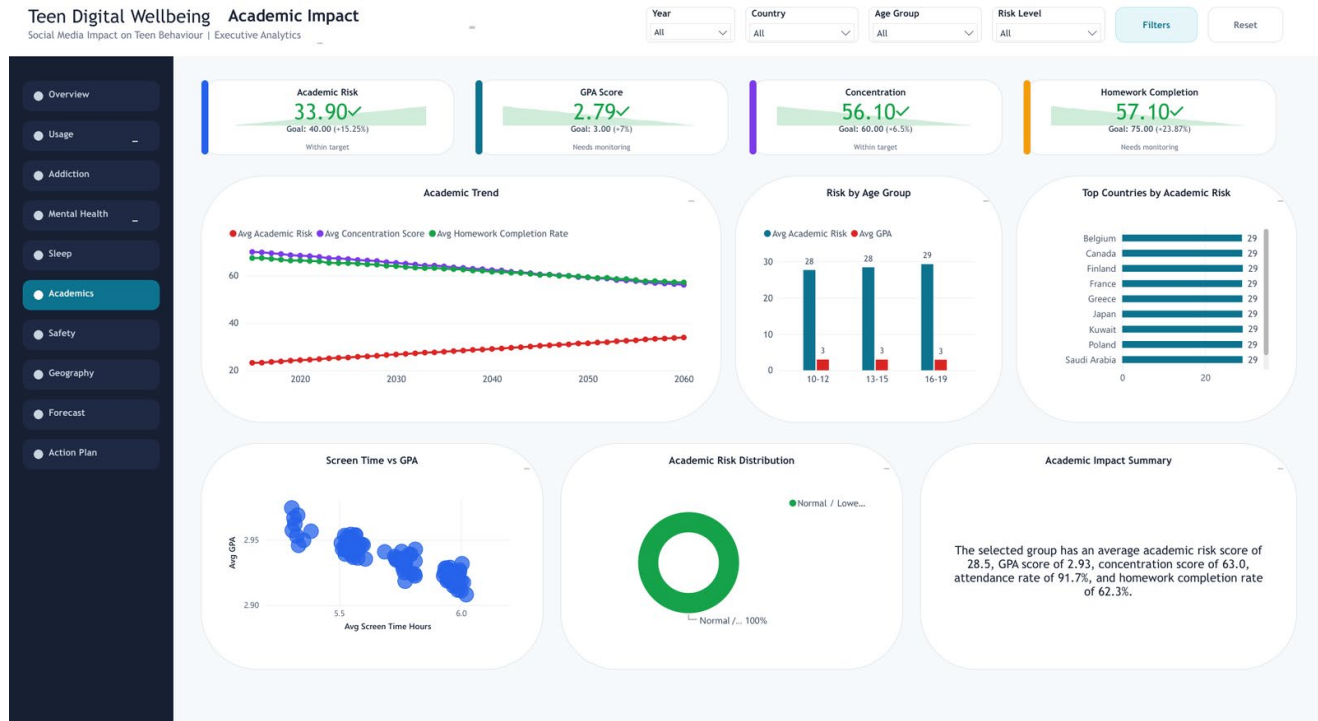


Figure 6: 06 Academic Impact page screenshot

07 Online Safety

Monitors cyberbullying, harmful content, misinformation, online harassment, and privacy risks.

Item	Details
KPI row	Cyberbullying Exposure, Online Harassment, Harmful Content, Privacy Risk
Key visuals	Online Safety Trend, Safety by Age Group, Top Countries by Cyberbullying, Cyberbullying Distribution, Harmful Content vs Misinformation, Online Safety Summary
Business use	Use this page for online safety training, policy monitoring, and digital citizenship programs.

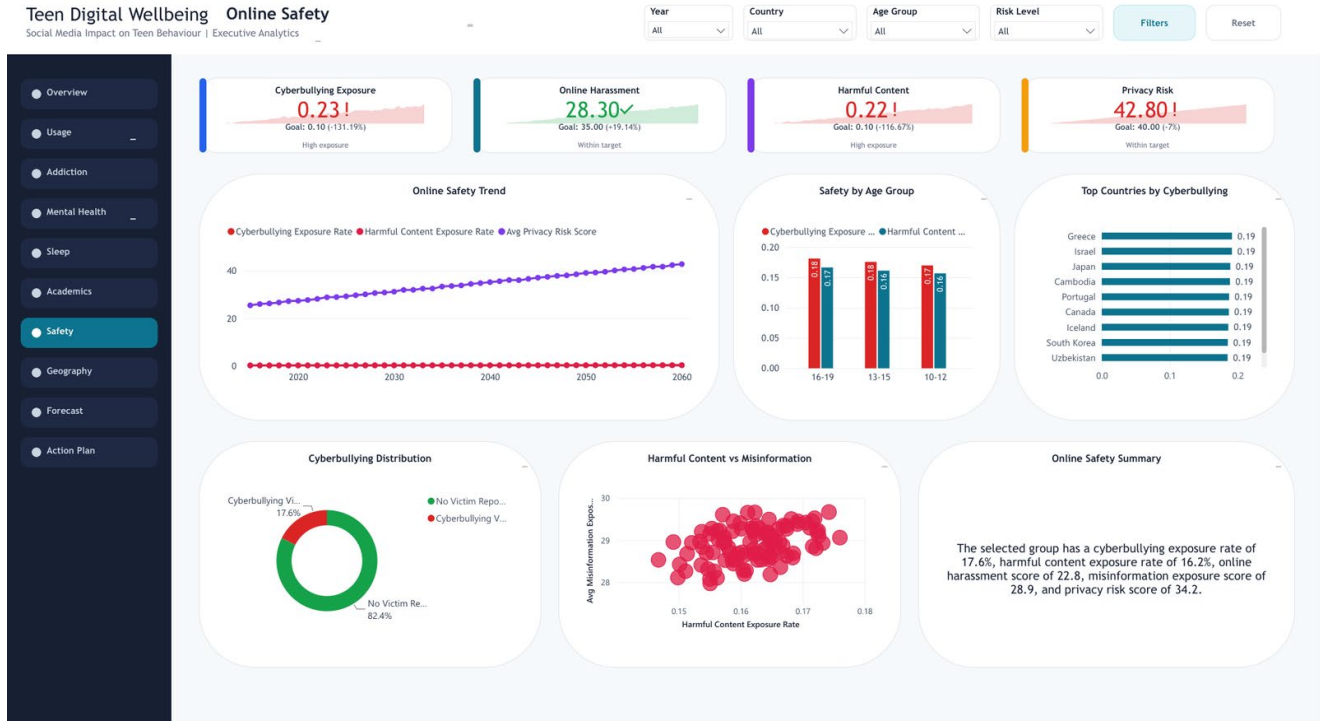
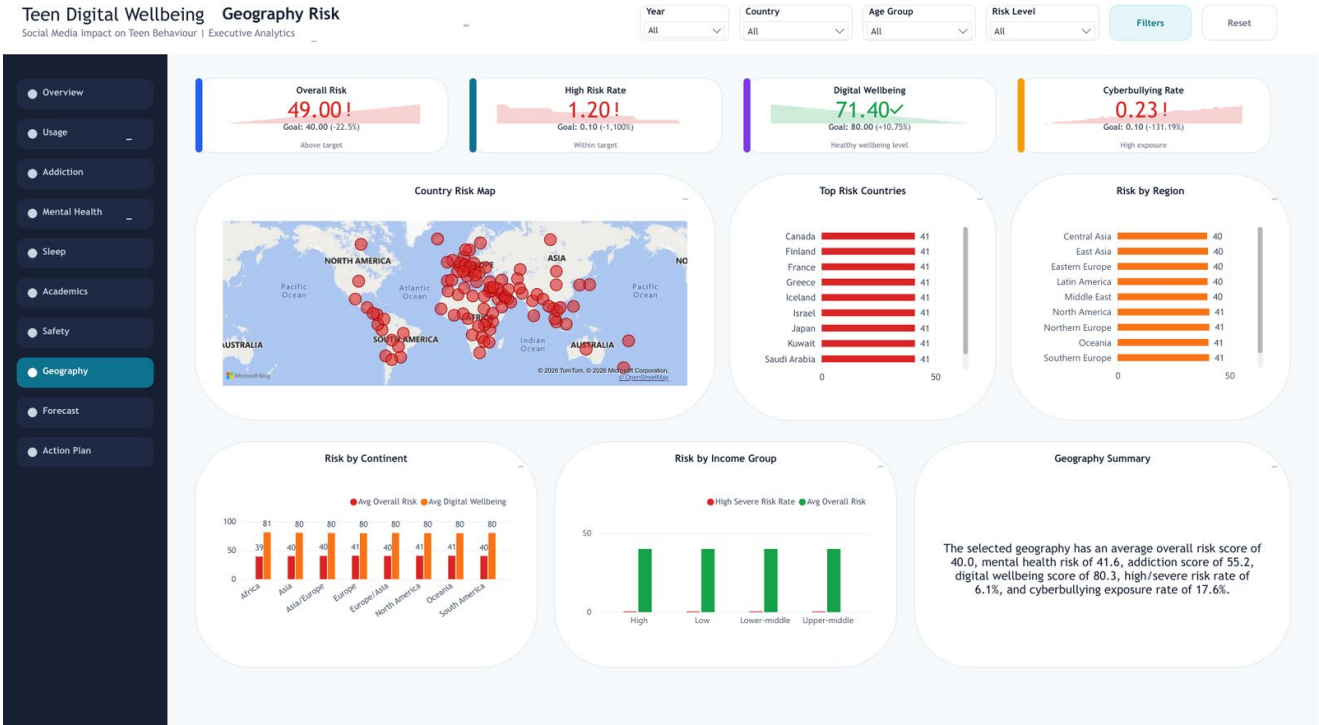


Figure 7: 07 Online Safety page screenshot

08 Geography Risk

Compares overall risk by country, region, continent, income group, and geography-related segments.

Item	Details
KPI row	Overall Risk, High Risk Rate, Digital Wellbeing, Cyberbullying Rate
Key visuals	Country Risk Map, Top Risk Countries, Risk by Region, Risk by Continent, Risk by Income Group, Geography Summary
Business use	Use this page to prioritize countries, regions, and income groups for targeted analysis.



09 Forecast Analysis

Projects addiction risk and mental health risk for 2030, 2040, 2050, and 2060.

Item	Details
KPI row	Addiction Risk 2060, Mental Risk 2060, Addiction Change, Mental Change
Key visuals	Forecast Risk Trend, Top Countries by 2060 Risk, Forecast by Region, Forecast by Income Group, Addiction vs Mental Forecast, Forecast Summary
Business use	Use this page to support forward-looking planning and long-term risk reduction strategy.

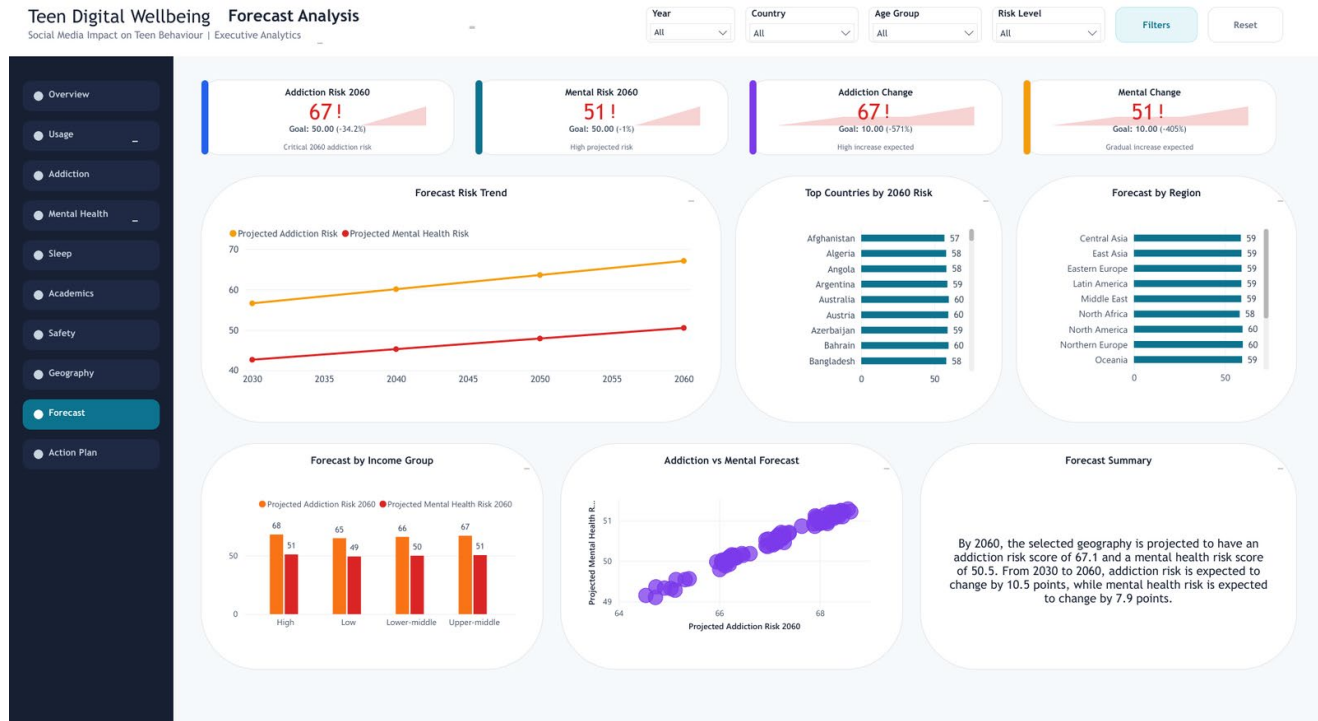


Figure 9: 09 Forecast Analysis page screenshot

12 Action Plan

Converts analysis into intervention categories, priority groups, country actions, and execution matrix.

Item	Details
KPI row	Immediate Intervention, High Priority Monitoring, Actionable Group, Avg Action Risk
Key visuals	Action Priority Distribution, Top Intervention Countries, Intervention Area Breakdown, Priority by Age Group, Action Matrix, Action Plan Summary
Business use	Use this page as the final decision-making page for operational action planning.



7. Power BI Service Deployment Runbook

The report should be published into DEV first. STAGE and PROD should be populated through the deployment pipeline where possible. The app should be created only from PROD.

Stage	Workspace	Purpose
DEV	Teen Digital Wellbeing - DEV	Development workspace. Publish the PBIX here first and validate report + semantic model.
STAGE	Teen Digital Wellbeing - STAGE / Test	Testing workspace. Validate deployed report, refresh, navigation, and business logic.
PROD	Teen Digital Wellbeing - PROD	Production workspace. Source for the Power BI App.

7.1 DEV publish process

1. Open the final PBIX in Power BI Desktop.
2. Validate that only pages 01-10 are visible.
3. Run Home > Refresh and confirm the report works in Desktop.
4. Publish to the DEV workspace only.
5. Open the DEV workspace in Power BI Service and validate both the report and semantic model are present.

8. Deployment Pipeline: DEV to STAGE to PROD

Use deployment pipelines for controlled release. Always deploy the report and semantic model together using Select related. Do not deploy only the report.

Activity	Instruction
Create pipeline	Create a pipeline named Teen Digital Wellbeing Deployment Pipeline.
Assign workspaces	Assign DEV, STAGE, and PROD workspaces to the corresponding pipeline stages.
Deploy DEV to STAGE	Select report, click Select related to include the semantic model, uncheck Continue deployment if items fail, then deploy.
Validate STAGE	Open the Stage report and test all pages, slicers, navigation, reset buttons, and forecast/action pages.
Deploy STAGE to PROD	Repeat the same process from Stage to Production. Deploy report and semantic model together.
Update app	Create or update the app only from the PROD workspace.
Common issue	Resolution
Visuals break after deployment but work in Desktop/DEV	Delete old report and semantic model from STAGE, then redeploy from DEV using Select related. Avoid partial deployments.
Semantic model warning icons in Service	Check whether report visuals work. If refresh succeeds and visuals work, warnings may be non-blocking. If visuals break, fix in Desktop and republish to DEV.
Semantic model not added to app	This is expected. Add the report to the app; the semantic model stays behind the report. Build permission is only needed for users creating new reports.
Duplicate reports/models in stage	Remove duplicates before deployment. Keep one report and one semantic model with aligned names.
Data source or parameter rules break deployment	Disable deployment rules unless stage-specific sources are required.

9. Power BI App Distribution

Setting	Recommended Value
App source workspace	PROD workspace only
App name	Teen Digital Wellbeing Analytics
Audience	Portfolio / Recruiter View or relevant internal team group
Content	Add the final Power BI report only. Do not manually add semantic model.
Build permission	Off for viewers/recruiters; On only for BI developers who need to build new reports.
Share permission	Off unless business policy allows resharing.
Update process	After deployment to PROD, open PROD workspace and click Update app.

10. QA, Validation, and Troubleshooting

Validation Area	Checklist
Page scope	Only pages 01-10 are visible. No removed pages appear in Service or App.
Slicers	Year, Country, Age Group, Risk Level work on applicable pages. Forecast page behavior is understood because forecast table may not respond to age/risk slicers.
Navigation	Left navigation buttons open correct pages. Active page highlight is correct.
Reset buttons	Reset bookmark clears slicers as expected. Bookmark data/display/current page options are configured.
Semantic model	Refresh succeeds in DEV/STAGE/PROD. Report is bound to correct semantic model.
Performance	Pages load in acceptable time. No excessive scrollbars or visual-level errors.
App	App opens as a viewer. Build permission is not accidentally granted to viewers.

11. Maintenance and Ownership

Role	Responsibilities
Power BI Developer	Maintain PBIX, DAX measures, report pages, visual formatting, bookmarks, and deployment pipeline.
BI Admin	Manage workspace access, app audience, gateway/credentials if required, and tenant-level settings.
Data Owner	Validate source definitions, risk logic, business thresholds, and action categories.
QA Reviewer	Validate report behavior in Stage before production release.
Business Stakeholder	Approve changes to KPIs, targets, interpretation, and action plan framework.
Change Type	Recommended process
Visual layout change	Modify PBIX in Desktop, publish to DEV, deploy to STAGE, validate, deploy to PROD, update app.
DAX measure change	Validate measure in Desktop and DEV before pipeline deployment.
Data source change	Update Power Query and credentials in DEV, then configure deployment rules if source differs by stage.
New page addition	Add to navigation and app only after business approval; update this documentation.
App audience change	Make change in PROD app audience settings and notify affected users.

12. Appendix: Key Measures and Operational Checklists

12.1 Page 10 Action Plan measures

```

Immediate Intervention Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[Action Priority] = "Immediate Intervention"
)

Immediate Intervention Rate =
DIVIDE ( [Immediate Intervention Records], [Teen Records] )

Actionable Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[Action Priority]
        IN { "Immediate Intervention", "High Priority Monitoring", "Preventive Support" }
)

Actionable Rate = DIVIDE ( [Actionable Records], [Teen Records] )

```

12.2 Deployment rescue checklist

- Confirm Desktop and DEV report are working.

7. Delete broken report and semantic model from STAGE if safe.
8. In the pipeline, select the report and use Select related to include the semantic model.
9. Uncheck Continue deployment in case one or more items fail.
10. Deploy DEV to STAGE and validate STAGE report pages.
11. If STAGE works, repeat clean deployment to PROD.
12. Update or create the app from PROD only.

12.3 Service data add guidance

Scenario	Recommended approach
Add full PBIX report and data	Use Power BI Desktop publish or Service workspace upload. This creates both report and semantic model.
Add simple Excel/CSV source	Use Workspace > New item > Semantic model or upload file; suitable for simple datasets.
Build report using existing model	Create report from existing semantic model; does not copy the model.
Move model to another workspace	Use deployment pipeline or republish PBIX to target workspace.

Final Notes

This documentation should be stored with the PBIX, README, screenshots, and any data/source files used for the project. When a production change is made, update this document, validate in STAGE, and publish the updated app from PROD only.

13. Appendix: Expanded DAX Measure Catalog

This appendix expands the DAX library used across the report pages. Keep these measures in the dedicated _Measures table where possible. Calculated columns should remain in their relevant table, especially when used in slicers, legends, matrix columns, or donut chart categories.

13.1 Core record, risk, and KPI measures

```

Teen Records =
COUNTROWS ( FactTeenBehaviour )

Avg Overall Risk =
AVERAGE ( FactTeenBehaviour[overall_risk_score] )

Avg Digital Wellbeing =
AVERAGE ( FactTeenBehaviour[digital_wellbeing_score] )

High Severe Risk Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[risk_level] IN { "High", "Severe" }
)

High Severe Risk Rate =
DIVIDE ( [High Severe Risk Records], [Teen Records] )

Problematic Use Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[problematic_use_flag] = 1
)

Problematic Use Rate =
DIVIDE ( [Problematic Use Records], [Teen Records] )

Cyberbullying Exposure Records =
CALCULATE (

```

Teen Digital Wellbeing Analytics - Team Documentation

```
[Teen Records],
FactTeenBehaviour[cyberbullying_victim] = 1
)

Cyberbullying Exposure Rate =
DIVIDE ( [Cyberbullying Exposure Records], [Teen Records] )

Avg Overall Risk KPI =
ROUND ( [Avg Overall Risk], 1 )

High Severe Risk Rate KPI =
[High Severe Risk Rate]

Avg Digital Wellbeing KPI =
ROUND ( [Avg Digital Wellbeing], 1 )
```

13.2 Executive summary support measures

```
Overall Risk Target =
40.0

Digital Wellbeing Target =
80.0

Executive Summary Text =
"The selected view contains "
& FORMAT ( [Teen Records], "#,##0" )
& " teen records. Average daily screen time is "
& FORMAT ( [Avg Screen Time Hours], "0.0" )
& " hours, average social media usage is "
& FORMAT ( [Avg Social Media Hours], "0.0" )
& " hours, average mental health risk is "
& FORMAT ( [Avg Mental Health Risk], "0.0" )
& ", and the high/severe risk rate is "
& FORMAT ( [High Severe Risk Rate], "0.0%" )
& ". Problematic use is "
& FORMAT ( [Problematic Use Rate], "0.0%" )
& " and cyberbullying exposure is "
& FORMAT ( [Cyberbullying Exposure Rate], "0.0%" )
& "."

Overall Risk KPI Status =
SWITCH (
    TRUE (),
    [Avg Overall Risk] >= 60, "Critical risk",
    [Avg Overall Risk] >= 45, "High risk",
    [Avg Overall Risk] > [Overall Risk Target], "Above target",
    [Avg Overall Risk] <= [Overall Risk Target], "Within target",
    "No data"
)

Digital Wellbeing KPI Status =
SWITCH (
    TRUE (),
    [Avg Digital Wellbeing] >= [Digital Wellbeing Target], "Healthy wellbeing level",
    [Avg Digital Wellbeing] >= 70, "Moderate wellbeing",
    [Avg Digital Wellbeing] >= 60, "Needs monitoring",
    [Avg Digital Wellbeing] < 60, "Low wellbeing",
    "No data"
)
```

13.3 Usage analysis measures

```
Avg Screen Time Hours =
AVERAGE ( FactTeenBehaviour[daily_screen_time_hours] )

Avg Social Media Hours =
AVERAGE ( FactTeenBehaviour[social_media_hours] )

Avg Night Screen Time Hours =
AVERAGE ( FactTeenBehaviour[night_screen_time_hours] )

Avg Daily Notifications =
AVERAGE ( FactTeenBehaviour[daily_notifications] )

Avg Educational Screen Hours =
AVERAGE ( FactTeenBehaviour[educational_screen_hours] )

Avg Gaming Hours =
AVERAGE ( FactTeenBehaviour[gaming_hours] )

Avg Platform Hours =
AVERAGE ( FactPlatformUsage[platform_hours] )

Avg Platform Hours Among Users =
DIVIDE (
    SUM ( FactPlatformUsage[platform_hours] ),
    CALCULATE (
        DISTINCTCOUNT ( FactPlatformUsage[record_id] ),
        FactPlatformUsage[platform_hours] > 0
    )
)

Usage Page Summary =
"The selected group spends an average of "
& FORMAT ( [Avg Screen Time Hours], "0.0" )
& " hours per day on screens, including "
& FORMAT ( [Avg Social Media Hours], "0.0" )
& " hours on social media and "
& FORMAT ( [Avg Night Screen Time Hours], "0.0" )
& " hours at night. Average daily notifications are "
& FORMAT ( [Avg Daily Notifications], "0" )
& "."
```

13.4 Addiction dependency measures

```
Avg Addiction Score =
AVERAGE ( FactTeenBehaviour[social_media_addiction_score] )

Avg Smartphone Dependency Score =
AVERAGE ( FactTeenBehaviour[smartphone_dependency_score] )

Avg FOMO Score =
AVERAGE ( FactTeenBehaviour[fear_of_missing_out_score] )

Avg Doomscrolling Score =
AVERAGE ( FactTeenBehaviour[doomscrolling_score] )

Avg Addiction Score KPI =
ROUND ( [Avg Addiction Score], 1 )

Avg Smartphone Dependency KPI =
ROUND ( [Avg Smartphone Dependency Score], 1 )

Avg FOMO Score KPI =
ROUND ( [Avg FOMO Score], 1 )
```

```
Problematic Use Rate KPI =  
[Problematic Use Rate]  
  
Addiction Page Summary =  
"The selected group has an average addiction score of "  
& FORMAT ( [Avg Addiction Score], "0.0" )  
& ", smartphone dependency score of "  
& FORMAT ( [Avg Smartphone Dependency Score], "0.0" )  
& ", FOMO score of "  
& FORMAT ( [Avg FOMO Score], "0.0" )  
& ", and problematic use rate of "  
& FORMAT ( [Problematic Use Rate], "0.0%" )  
& "."
```

13.5 Mental health measures

```
Avg Mental Health Risk =  
AVERAGE ( FactTeenBehaviour[mental_health_risk_score] )  
  
Avg Anxiety Score =  
AVERAGE ( FactTeenBehaviour[anxiety_score] )  
  
Avg Depression Score =  
AVERAGE ( FactTeenBehaviour[depression_score] )  
  
Avg Stress Score =  
AVERAGE ( FactTeenBehaviour[stress_score] )  
  
Avg Loneliness Score =  
AVERAGE ( FactTeenBehaviour[loneliness_score] )  
  
Avg Mental Health Risk KPI =  
ROUND ( [Avg Mental Health Risk], 1 )  
  
Avg Anxiety Score KPI =  
ROUND ( [Avg Anxiety Score], 1 )  
  
Avg Depression Score KPI =  
ROUND ( [Avg Depression Score], 1 )  
  
Avg Stress Score KPI =  
ROUND ( [Avg Stress Score], 1 )  
  
Mental Health Page Summary =  
"The selected group has an average mental health risk score of "  
& FORMAT ( [Avg Mental Health Risk], "0.0" )  
& ", anxiety score of "  
& FORMAT ( [Avg Anxiety Score], "0.0" )  
& ", depression score of "  
& FORMAT ( [Avg Depression Score], "0.0" )  
& ", stress score of "  
& FORMAT ( [Avg Stress Score], "0.0" )  
& ", and loneliness score of "  
& FORMAT ( [Avg Loneliness Score], "0.0" )  
& "."
```

13.6 Sleep, physical wellbeing, and academic impact measures

```
Avg Sleep Hours =  
AVERAGE ( FactTeenBehaviour[sleep_hours] )  
  
Avg Physical Activity Hours =  
AVERAGE ( FactTeenBehaviour[physical_activity_hours] )
```

```
Avg Outdoor Activity Hours =  
AVERAGE ( FactTeenBehaviour[outdoor_activity_hours] )  
  
Avg Eye Strain Score =  
AVERAGE ( FactTeenBehaviour[eye_strain_score] )  
  
Avg Headache Frequency =  
AVERAGE ( FactTeenBehaviour[headache_frequency_weekly] )  
  
Avg Physical Health Risk =  
AVERAGE ( FactTeenBehaviour[physical_health_risk_score] )  
  
Avg Academic Risk =  
AVERAGE ( FactTeenBehaviour[academic_risk_score] )  
  
Avg GPA =  
AVERAGE ( FactTeenBehaviour[gpa_score] )  
  
Avg Concentration Score =  
AVERAGE ( FactTeenBehaviour[concentration_score] )  
  
Avg Attendance Rate =  
AVERAGE ( FactTeenBehaviour[school_attendance_rate_pct] )  
  
Avg Homework Completion Rate =  
AVERAGE ( FactTeenBehaviour[homework_completion_rate_pct] )  
  
High Academic Risk Records =  
CALCULATE (  
    [Teen Records],  
    FactTeenBehaviour[high_academic_risk_flag] = 1  
)  
  
High Academic Risk Rate =  
DIVIDE ( [High Academic Risk Records], [Teen Records] )
```

13.7 Online safety measures

```
Avg Online Harassment Score =  
AVERAGE ( FactTeenBehaviour[online_harassment_score] )  
  
Avg Harmful Content Exposure =  
AVERAGE ( FactTeenBehaviour[harmful_content_exposure] )  
  
Avg Misinformation Exposure Score =  
AVERAGE ( FactTeenBehaviour[misinformation_exposure_score] )  
  
Avg Privacy Risk Score =  
AVERAGE ( FactTeenBehaviour[privacy_risk_score] )  
  
Cyberbullying Offender Records =  
CALCULATE (  
    [Teen Records],  
    FactTeenBehaviour[cyberbullying_offender] = 1  
)  
  
Cyberbullying Offender Rate =  
DIVIDE ( [Cyberbullying Offender Records], [Teen Records] )  
  
Harmful Content Exposure Records =  
CALCULATE (  
    [Teen Records],  
    FactTeenBehaviour[harmful_content_exposure] = 1  
)
```

Teen Digital Wellbeing Analytics - Team Documentation

```
Harmful Content Exposure Rate =  
DIVIDE ( [Harmful Content Exposure Records], [Teen Records] )
```

```
Online Safety Page Summary =  
"The selected group has a cyberbullying exposure rate of "  
& FORMAT ( [Cyberbullying Exposure Rate], "0.0%" )  
& ", harmful content exposure rate of "  
& FORMAT ( [Harmful Content Exposure Rate], "0.0%" )  
& ", online harassment score of "  
& FORMAT ( [Avg Online Harassment Score], "0.0" )  
& ", misinformation exposure score of "  
& FORMAT ( [Avg Misinformation Exposure Score], "0.0" )  
& ", and privacy risk score of "  
& FORMAT ( [Avg Privacy Risk Score], "0.0" )  
& "."
```

13.8 Geography and ranking measures

```
Country Risk Rank =  
RANKX (   
    ALLSELECTED ( DimCountry[Country] ),  
    [Avg Overall Risk],  
    ,  
    DESC,  
    DENSE  
)
```

```
Show Top 10 Risk Countries =  
IF ( [Country Risk Rank] <= 10, 1, 0 )
```

```
Region Risk Rank =  
RANKX (   
    ALLSELECTED ( DimCountry[Region] ),  
    [Avg Overall Risk],  
    ,  
    DESC,  
    DENSE  
)
```

```
Show Top 10 Risk Regions =  
IF ( [Region Risk Rank] <= 10, 1, 0 )
```

```
Geography Risk Page Summary =  
"The selected geography has an average overall risk score of "  
& FORMAT ( [Avg Overall Risk], "0.0" )  
& ", mental health risk of "  
& FORMAT ( [Avg Mental Health Risk], "0.0" )  
& ", addiction score of "  
& FORMAT ( [Avg Addiction Score], "0.0" )  
& ", digital wellbeing score of "  
& FORMAT ( [Avg Digital Wellbeing], "0.0" )  
& ", high/severe risk rate of "  
& FORMAT ( [High Severe Risk Rate], "0.0%" )  
& ", and cyberbullying exposure rate of "  
& FORMAT ( [Cyberbullying Exposure Rate], "0.0%" )  
& "."
```

13.9 Forecast analysis measures

```
Projected Risk =  
AVERAGE ( FactForecastLong[projected_risk] )
```

```
Projected Addiction Risk =  
CALCULATE (
```

```

    [Projected Risk],
    FactForecastLong[risk_type] = "Addiction Risk"
)

Projected Mental Health Risk =
CALCULATE (
    [Projected Risk],
    FactForecastLong[risk_type] = "Mental Health Risk"
)

Projected Addiction Risk 2030 =
CALCULATE (
    [Projected Addiction Risk],
    DimForecastYear[forecast_year] = 2030
)

Projected Addiction Risk 2060 =
CALCULATE (
    [Projected Addiction Risk],
    DimForecastYear[forecast_year] = 2060
)

Projected Mental Health Risk 2030 =
CALCULATE (
    [Projected Mental Health Risk],
    DimForecastYear[forecast_year] = 2030
)

Projected Mental Health Risk 2060 =
CALCULATE (
    [Projected Mental Health Risk],
    DimForecastYear[forecast_year] = 2060
)

Addiction Forecast Change 2030 to 2060 =
[Projected Addiction Risk 2060] - [Projected Addiction Risk 2030]

Mental Health Forecast Change 2030 to 2060 =
[Projected Mental Health Risk 2060] - [Projected Mental Health Risk 2030]

Projected Combined Risk 2060 =
CALCULATE (
    [Projected Risk],
    DimForecastYear[forecast_year] = 2060
)

Forecast Page Summary =
"By 2060, the selected geography is projected to have an addiction risk score of "
& FORMAT ( [Projected Addiction Risk 2060], "0.0" )
& " and a mental health risk score of "
& FORMAT ( [Projected Mental Health Risk 2060], "0.0" )
& ". From 2030 to 2060, addiction risk is expected to change by "
& FORMAT ( [Addiction Forecast Change 2030 to 2060], "0.0" )
& " points, while mental health risk is expected to change by "
& FORMAT ( [Mental Health Forecast Change 2030 to 2060], "0.0" )
& " points."

```

13.10 Action plan calculated columns and measures

The following items support Page 10 and must be created as calculated columns when used in legends, slicers, matrix columns, or donut categories.

```

Action Priority =
SWITCH (
    TRUE (),

```


Teen Digital Wellbeing Analytics - Team Documentation

```

FactTeenBehaviour[risk_level] = "Severe"
    || FactTeenBehaviour[overall_risk_score] >= 70,
    "Immediate Intervention",
FactTeenBehaviour[risk_level] = "High"
    || FactTeenBehaviour[overall_risk_score] >= 55,
    "High Priority Monitoring",
FactTeenBehaviour[risk_level] = "Medium"
    || FactTeenBehaviour[overall_risk_score] >= 40,
    "Preventive Support",
    "General Awareness"
)

Primary Intervention Area =
SWITCH (
    TRUE (),
    FactTeenBehaviour[high_mental_health_risk_flag] = 1, "Mental Health Support",
    FactTeenBehaviour[high_academic_risk_flag] = 1, "Academic Support",
    FactTeenBehaviour[cyberbullying_victim] = 1, "Online Safety Support",
    FactTeenBehaviour[problematic_use_flag] = 1, "Digital Use Coaching",
    FactTeenBehaviour[sleep_hours] < 7, "Sleep & Physical Wellbeing",
    FactTeenBehaviour[daily_screen_time_hours] >= 7, "Screen Time Reduction",
    "General Awareness"
)

Immediate Intervention Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[Action Priority] = "Immediate Intervention"
)

Immediate Intervention Rate =
DIVIDE ( [Immediate Intervention Records], [Teen Records] )

High Priority Monitoring Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[Action Priority] = "High Priority Monitoring"
)

High Priority Monitoring Rate =
DIVIDE ( [High Priority Monitoring Records], [Teen Records] )

Actionable Records =
CALCULATE (
    [Teen Records],
    FactTeenBehaviour[Action Priority]
        IN { "Immediate Intervention", "High Priority Monitoring", "Preventive Support" }
)

Actionable Rate =
DIVIDE ( [Actionable Records], [Teen Records] )

Action Plan Summary =
"The selected population has "
& FORMAT ( [Immediate Intervention Rate], "0.0%" )
& " of teens needing immediate intervention, "
& FORMAT ( [High Priority Monitoring Rate], "0.0%" )
& " requiring high-priority monitoring, and "
& FORMAT ( [Actionable Rate], "0.0%" )
& " falling into an actionable support group. The average overall risk score is "
& FORMAT ( [Avg Overall Risk], "0.0" )
& ". Recommended focus areas are mental health support, digital use coaching, sleep wellbeing, academic
support, and online safety."

```

13.11 Data quality and governance helper measures

These optional measures can be used if the team later decides to reintroduce a governance page or add a hidden QA page for semantic model validation.

```
Total Records =
COUNTROWS ( FactTeenBehaviour )

Total Countries =
DISTINCTCOUNT ( FactTeenBehaviour[country] )

Total Years =
DISTINCTCOUNT ( FactTeenBehaviour[year] )

Missing Overall Risk Records =
CALCULATE (
    [Total Records],
    ISBLANK ( FactTeenBehaviour[overall_risk_score] )
)

Missing Screen Time Records =
CALCULATE (
    [Total Records],
    ISBLANK ( FactTeenBehaviour[daily_screen_time_hours] )
)

Missing Sleep Records =
CALCULATE (
    [Total Records],
    ISBLANK ( FactTeenBehaviour[sleep_hours] )
)

Missing Critical Data Records =
[Missing Overall Risk Records]
+ [Missing Screen Time Records]
+ [Missing Sleep Records]

Missing Critical Data Rate =
DIVIDE ( [Missing Critical Data Records], [Total Records] )

Data Completeness Rate =
1 - [Missing Critical Data Rate]

Data Quality Status =
SWITCH (
    TRUE (),
    [Data Completeness Rate] >= 0.98, "Reporting ready",
    [Data Completeness Rate] >= 0.95, "Minor quality issues",
    [Data Completeness Rate] >= 0.90, "Needs validation",
    "Data quality risk"
)
```

13.12 DAX formatting and governance rules

- Keep all reusable measures in the _Measures table and organize them by display folders such as Executive, Usage, Addiction, Mental Health, Forecast, and Action Plan.
- Use measures for aggregations and KPIs. Use calculated columns only when the result is required as a slicer, legend, matrix row/column, relationship key, or static category.
- Format rate measures as Percentage with one decimal place. Format score measures as Decimal number with one decimal place. Format record counts with whole number and thousand separators.
- Avoid using fact-to-fact relationships. Use shared dimensions such as DimDate, DimCountry, DimIncomeGroup, DimRisk, DimUrbanRural, and DimForecastYear to control filters.

- For Deployment Pipelines, validate the semantic model and report together in DEV before deploying to STAGE and PROD.